

MATH102 SMP Week 3

2020F Mid Problem 1

Find a parametric equation for the line of intersection of two planes in $\mathbb{R}^3$ given by equations $2x - y + 3z = 5$ and $-x + y - z = 1$.

2020F Mid Problem 2

Two lines ℓ_1 and ℓ_2 in $\mathbb{R}^3$ are given as follows: The line ℓ_1 is parallel to the vector $(1, 1, 1)$ and passes through the point $(0, 0, 0)$, whereas the other line ℓ_2 is parallel to the vector $(1, -1, 0)$ passing through $(0, 0, 1)$. Now, find the (shortest) distance d between ℓ_1 and ℓ_2 .

Recall: Continuous on a single-valued function

Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as follow:

$$f(x) = \begin{cases} 1 & (x = 0) \\ \frac{\sin x}{x} & (\text{otherwise}). \end{cases}$$

Show that f is continuous at $x = 0$ by using definition of $\epsilon - \delta$ argument.
(Hint: $\sin x < x < \tan x$ if $x \in \mathbb{R}^+$)

Textbook 1.63

Let $W_1 = (1, 1, 1, 0)$ and $W_2 = (0, 1, 1, 1)$. Find two linearly independent vectors that are orthogonal to both W_1 and W_2 .

Textbook 1.95

Find equations in the form $ax + by + cz = d$ for the following planes in $\mathbb{R}^3$.

- (a) The plane through the origin with normal $(1, 0, 0)$.
- (b) The plane through $(0, 0, 0)$, $(0, 1, 1)$, and $(-3, 0, 0)$.
- (c) The plane containing the point $(1, 1, 1)$ and parallel to the plane with equation $x - 3y + 5z = 60$.

Textbook 2.4

Show that a constant function $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear if and only if $F(x_1, x_2, \dots, x_n) = 0$.

Textbook 2.35

Show that the set $x^2 + y^2 < 1$ in $\mathbb{R}^3$ is an open set.

Textbook 2.36

Let $S = \mathbb{R}^2 \setminus \{0\}$. Show that 0 is a boundary point of S .